

MARIAN COLLEGE KUTTIKKANAM (AUTONOMOUS)
UG Programme (Honors) Degree October 2025 Regular Examinations
First Semester

251SECUBC101 Fundamentals of Programming Using C++

Time: 2 Hours

Maximum Marks: 40

Answer one question from each bunch

BUNCH I

- I.A You are developing a program to manage sales data for a transaction. The program should perform the following tasks:
- Input the product's name, quantity sold, and price per unit.
 - Calculate the revenue generated:
 - $\text{Revenue} = \text{Quantity Sold} \times \text{Price Per Unit}$.
 - Classify the sales performance based on revenue:
 - "Low" for revenue < 500, "Medium" for 500–1000, and "High" for revenue > 1000.
 - Display the information of the transaction, including the product name, quantity sold, price per unit, revenue, and performance category.

Draw a flowchart and write a step-by-step algorithm.

(CO1:8 Marks)

OR

- I.B ✓ You are developing a program called "Weather Report Classifier" to categorize daily weather conditions based on temperature readings.

- Input: City Name, Temperature (in °Celsius)

- If Temperature $< 15^{\circ}\text{C}$ $\rightarrow$ Display "Cold"
 - If Temperature $\geq 15^{\circ}\text{C}$ and $\leq 30^{\circ}\text{C}$ $\rightarrow$ Display "Moderate"
 - If Temperature $> 30^{\circ}\text{C}$ $\rightarrow$ Display "Hot"
 - Display City Name, Temperature, and Weather Category
- Draw a Flowchart and Write an Algorithm to illustrate the working of this program. (CO1:8 Marks)

BUNCH II

- II.A ✓ A software company is deciding between C++ and other programming languages for a new project. In this context, explain the history and evolution of C++ as a programming language. Discuss any two merits of object oriented programming concepts. (CO2:8 Marks)

OR

- II.B ✓ While reviewing a colleague's code, you notice that some parts of the code are difficult to understand. To help with readability, you decide to break down the code into its core components. In this context, explain what tokens are in C++ and provide examples of different types of tokens such as keywords, identifiers, constants, and operators. Discuss how each of these tokens is used in the code and how they form the building blocks of a C++ program. (CO2:8 Marks)

BUNCH III

- III.A A car rental service charges fuel costs based on the distance travelled:
- For the first 100 km: Rs.10 per km
 - For the next 200 km: Rs.8 per km
 - Beyond 300 km: Rs.5 per k
- Additionally, a service charge of Rs.200 is added to all bills. Write a C++ program to input the customer's name, distance traveled,

and calculate the total charges. Display the customer's name and the total amount.
(CO3:8 Marks)

OR

- ✓ III.B A school offers scholarships based on marks:
- Marks ≥ 90 : Full scholarship (100% fee waiver)
 - Marks ≥ 75 and < 90 : 50% fee waiver
 - Marks ≥ 60 and < 75 : 25% fee waiver
 - Marks < 60 : No scholarship

Write a C++ program to input the student's name and marks. Display the student's name, marks, and the scholarship percentage.

(CO3:8 Marks)

BUNCH IV

✓ IV.A Define a class Vehicle with:

- Private members: model, makeYear, price.
- Use constructor overloading to initialize the attributes with different values.
- Public member function to display the details.
- Use a destructor to print a message as "Object Destroyed".

Explain the concepts with the complete C++ Program.

(CO4:8 Marks)

OR

IV.B Define a class ArrayManipulator with:

- **Private Data Members:** An array and its size.
- **Member Functions:**
 - A function to accept array elements. (Public)
 - A function to reverse and display the reversed array (Private)

Write the complete program and explain the concepts used.

(CO4:8 Marks)

BUNCH V

- ✓ V.A Write a C++ program that dynamically allocates memory for an array of n cities' populations (integer). The program should read the data, calculate and print the total population, and then deallocate the memory using memory management operator.
(CO5:8 Marks)

OR

- V.B Write a C++ program that dynamically allocates memory for an array of n months' rainfall data (float). The program should read the data, calculate and print the total rainfall for the year, and then deallocate the memory using delete operator.
(CO5:8 Marks)

CO1	Analyse the problem-solving process, focusing on problem analysis, design, programming languages, translators, and debugging.
CO2	Apply C++ fundamentals, structure, data types, operators, and basic input/output operations.
CO3	Apply decision-making, loops, and arrays in C++, including initialization, operations, and string manipulation.
CO4	Apply OOP concepts in C++ using classes, objects, access specifiers, constructors, destructors, and static members.
CO5	Implement C++ functions, recursion, pointers, dynamic memory, structures, and unions.